Electronics Workshop Module

Laboratory Safety Protocols & Electronic Components

Ateneo Computer Engineering Society

Department of Computer Engineering
College of Science, Engineering, and Architecture
Ateneo de Naga University

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Session Objectives & Scope

What You Will Learn Today

- **Laboratory Safety Compliance:** Recognize electrical physiological thresholds and shock hazards.
- **ESD Damage Prevention:** Static dissipation mechanisms and sensitive semiconductor handling.
- **Component Taxonomy:** Distinguish between passive and active electronic devices.
- **Packaging & Pinouts:** Read polarity markings, transistor pin assignments, and DIP IC indexing.

Workshop Roadmap

- **Part 1: Laboratory Safety Protocols**
  - Shock physics ($I = V/R_{\text{body}}$)
  - Physiological current thresholds
  - Workbench housekeeping & ESD control
- **Part 2: Electronic Components**
  - Passive components (R, C, L)
  - Active semiconductors (Diodes, LEDs, BJTs, ICs)
  - Essential circuit calculations

The Physics of Electric Shock: Human Body as a Resistor

Ohm's Law Applied to Human Tissue

$$I = \frac{V}{R_{\text{body}}}$$

- Voltage alone does not kill; **current magnitude (I)** and pathway dictate lethality.
- **Dry Skin:** $R \approx 10{,}000\,\Omega$ to $100{,}000\,\Omega$.
- **Wet/Perspiring Skin:** R plummets to $\leq 1{,}000\,\Omega$.
- At $1{,}000\,\Omega$, a $50\,\text{V}$ line yields $50\,\text{mA}$—enough to trigger ventricular fibrillation.

The "One-Hand Rule"

- Keep **one hand in your pocket** or behind your back when probing live high-voltage test points.
- Prevents current from forming a cross-torso circuit through the heart (hand-to-hand pathway).

Physiological Current Thresholds (60 Hz AC/DC)

Current Level	Physiological Reaction	Clinical Hazard
1 mA	Perception Threshold	Faint tingling sensation; safe.
5 mA	Maximum Harmless Current	Involuntary jerk; potential secondary trauma.
10–20 mA	**"Let-Go" Threshold**	Muscular paralysis; victim cannot release conductor.
20–50 mA	Respiratory Arrest	Chest muscles contract; suffocation within minutes.
$\geq$ 100 mA	**Ventricular Fibrillation**	Heart quivers uncontrollably; fatal without immediate CPR/AED.

Workbench Safety, Housekeeping & ESD Mitigation

Personal Protective Equipment (PPE)

- **ANSI Z87.1 Safety Glasses:** Mandatory during soldering and component lead clipping.
- **Zero Jewelry Policy:** Rings and metallic watchbands can short power rails, causing severe arc burns.
- **De-energized Prototyping:** Always disconnect power before inserting or pulling components.

Electrostatic Discharge (ESD) Standards

- Human perception threshold for ESD is $\approx 3{,}000\,\text{V}$.
- CMOS/MOSFET gates suffer punch-through breakdown at $< 100\,\text{V}$.
- **Wrist Straps:** Must include a $1\,\text{M}\Omega$ **safety resistor** to protect user from mains faults.
- Handle ICs by packaging body, never exposed pins.

Laboratory Power Tool Safety & Mechanical Hazards

PCB Drills, Rotary Tools & Heat Guns

- **Workpiece Clamping:** Always secure PCBs in a drill press vise or PCB clamp. *Never hand-hold a board while drilling.*
- **Power Disconnection:** Unplug or de-energize the power tool before changing drill bits, collets, or cutting discs.
- **Thermal Awareness:** Heat guns operate above 400°C. Rest solely on designated heat-resistant stands.

Entanglement & Dust Precautions

- **Zero Gloves Policy near Rotating Machinery:** Gloves create a lethal snagging hazard around rotating chucks and drill spindles.
- **Attire Standards:** Tie back long hair, tuck in loose clothing/lanyards, and roll up long sleeves.
- **Respiratory & Eye Protection:** Wear an N95 dust mask and ANSI Z87.1 eyewear when cutting/drilling FR4 fiberglass to prevent inhaling toxic particulates.

PCB Etching Chemicals: Safe Storage & Disposal Protocols

Safe Chemical Storage ($FeCl_3$ & Persulfates)

- **Container Compatibility:** Store strictly in heavy-duty, acid-resistant High-Density Polyethylene (HDPE) or polypropylene containers.
- **Zero Metal Contact:** Never use metallic containers or utensils; acidic etchants rapidly corrode metals and release gas.
- **Ventilated Storage:** Keep in dedicated corrosive cabinets away from organic solvents, bases, and heat.
- **Explicit Labeling:** Mark chemical name, concentration, preparation date, and hazard warning.

Disposal Protocols for Spent Etchant

- **Zero Sink Drainage:** Never pour spent etchant down domestic or municipal drains. Dissolved copper (Cu^{2+}) is biocidal and corrodes pipes.
- **Neutralization:** Slowly neutralize with sodium bicarbonate ($NaHCO_3$) until foaming stops.
- **Precipitation & Collection:** Allow copper hydroxide sludge to settle; collect in labeled hazardous waste carboys.
- **Institutional Transfer:** Transfer waste to University Environmental Safety for certified hazardous disposal.

Component Taxonomy: Passive vs. Active Devices

Passive Components

- Cannot introduce net electrical energy or amplify.
- **Resistors (R):** Dissipate energy as heat ($P = I^2R$).
- **Capacitors (C):** Store energy in electric field ($E = \frac{1}{2}CV^2$).
- **Inductors (L):** Store energy in magnetic field ($E = \frac{1}{2}LI^2$).

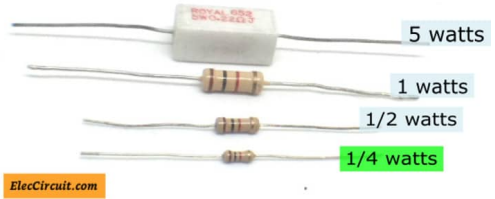
Active Components

- Require external DC bias; can amplify or switch.
- **Diodes / LEDs:** Unidirectional conduction ($V > V_F$).
- **BJTs:** Current-controlled current source ($I_C = \beta I_B$).
- **MOSFETs:** Voltage-controlled channel switching.
- **ICs:** Monolithic integration of micro-circuits.

Resistors: Color Codes & Power Ratings

Function & Sizing

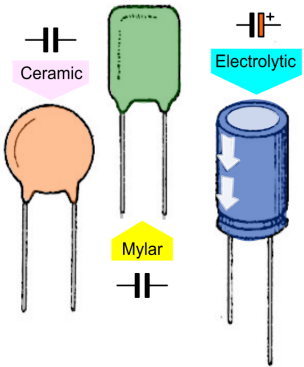
- Current limitation and potential division ($V = IR$).
- Physical size correlates with **thermal power dissipation**, not resistance value!
- Standard lab: 1/4 W carbon/metal film.
- High power: 1 W, 2 W, up to 5 W ceramic block.



Capacitors & Inductors: Energy Storage

Capacitor Types & Polarity Caution

- **Ceramic (Non-polarized):** High frequency bypass (10 pF–100 nF).
- **Electrolytic (Polarized):** High capacitance bulk filtering (1 μF–4700 μF).
- **Caution:** Reverse polarity on electrolytics causes catastrophic electrolyte boiling and venting!



3 most common capacitor types

Electrical Impedance (Z): Resistance in AC Circuits

Complex Impedance Definition

$$Z = R + jX = |Z|\angle\theta$$

- **Impedance (Z):** Generalized opposition to alternating current, incorporating both magnitude and phase shift.
- **Resistance (R):** Real component; irreversible power dissipation as heat ($P = I^2R$).
- **Reactance (X):** Imaginary component; reversible energy storage in electric or magnetic fields.
- Magnitude: $|Z| = \sqrt{R^2 + X^2}$.
- Phase Angle: $\theta = \arctan(X/R)$.

Frequency-Dependent Reactances ($\omega = 2\pi f$)

- **Capacitive Reactance (X_C):**

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} \implies Z_C = -j\frac{1}{\omega C}$$

Decreases as frequency rises. *Blocks DC ($f = 0$), passes high-frequency AC.*

- **Inductive Reactance (X_L):**

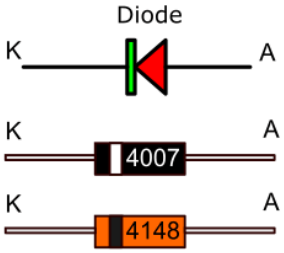
$$X_L = \omega L = 2\pi f L \implies Z_L = j\omega L$$

Increases as frequency rises. *Passes DC ($f = 0$), chokes high-frequency AC.*

Semiconductor Diodes & Cathode Band Identification

PN Junction Physics

- Acts as an electrical one-way check valve.
- **Forward Bias ($V > V_F$):** Conduction initiates ($V_F \approx 0.7\,\text{V}$ for Silicon, $0.3\,\text{V}$ for Schottky).
- **Reverse Bias:** Blocks current up to breakdown voltage V_{BR}.
- **Physical Marking:** Silver or black band denotes the **Cathode (K)** terminal.



Most Common Diodes

Light Emitting Diodes (LEDs) & Ballast Resistor Design

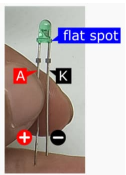
Current-Limiting Calculation

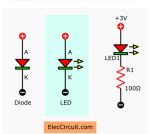
$$R = \frac{V_{CC} - V_F}{I_F}$$

Example (5V DC Supply, Standard Red LED):

$$R = \frac{5.0\,\text{V} - 2.0\,\text{V}}{0.0091\,\text{A}} \approx 330\,\Omega$$

- **Anode (+):** Longer lead.
- **Cathode (−):** Shorter lead & flat rim notch.





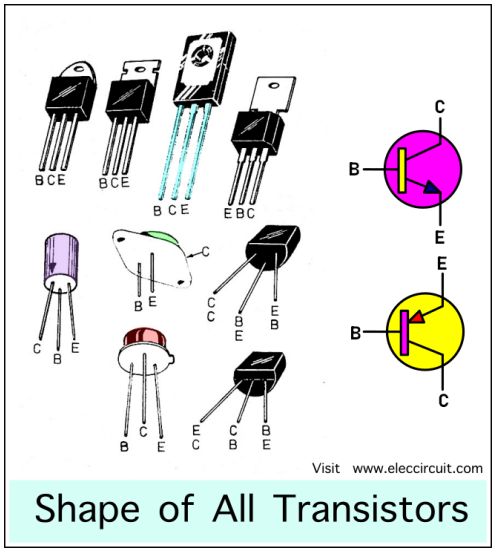
Bipolar Junction Transistors (BJTs) & Packaging

Three-Terminal Switching

- Terminals: **Base (B)**, **Collector (C)**, **Emitter (E)**.
- Small base current controls large collector current:

$$I_C = \beta I_B$$

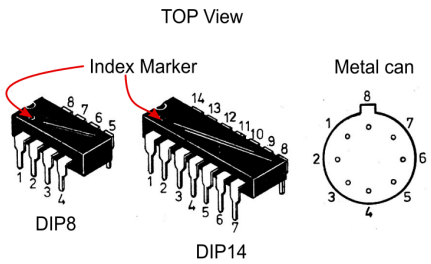
- **NPN:** Current flows Collector → Emitter when Base is positive.
- Always verify pinouts (TO-92 packages vary across manufacturers: E-B-C vs. C-B-E).



Integrated Circuits (ICs) & Dual In-Line Packaging (DIP)

Pin 1 Orientation Convention

- **Alignment Notch:** Position the notch facing upward or leftward.
- **Pin 1 Dot:** Directly adjacent to the notch.
- **Counter-Clockwise Counting:** Numbering proceeds down the left column and continues up the right column.
- Use IC extractors to avoid bending pins during removal.



Summary of Key Principles

Laboratory Safety Takeaways

- Current (I) dictates biological shock hazard ($\geq 10\,\text{mA}$ freezes muscles).
- Practice the **One-Hand Rule** around energized circuits.
- Always wear **ANSI Z87.1 eye protection**.
- Use grounded wrist straps with $1\,\textbf{M}\Omega$ **safety resistor** for ESD mitigation.

Component Engineering Takeaways

- Resistor physical volume reflects **power dissipation**, not ohms.
- Respect capacitor **polarity** to prevent explosive venting.
- Size series ballast resistors for LEDs using $R = (V_{CC} - V_F)/I_F$.
- Verify BJT pinouts (E-B-C) and locate IC **Pin 1** before powering.

Thank You!

Questions & Discussion

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